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PROGRAM TicTacToe

START Main
   CLEAR Screen
   DISPLAY Banner "TIC TAC TOE PROJECT"

   PROMPT "Player 1 Name: "
   READ n1
   TRIM spaces from n1
   IF n1 == "" THEN
      SET n1 = "Player 1"
   END IF
   IF length of n1 > 10 THEN
      SET n1 = first 10 characters of n1
   END IF

   PROMPT "Player 2 Name: "
   READ n2
   TRIM spaces from n2
   IF n2 == "" THEN
      SET n2 = "Player 2"
   END IF
   IF length of n2 > 10 THEN
      SET n2 = first 10 characters of n2
   END IF

   SET s1 = '1', s2 = '2', s3 = '3'
   SET s4 = '4', s5 = '5', s6 = '6'
   SET s7 = '7', s8 = '8', s9 = '9'

   SET game = 1
   SET scoreA = 0
   SET scoreB = 0

   WHILE game == 1 DO
      SET moves = 0
      SET gameOver = FALSE

      // Reset board slots
      SET s1 = '1', s2 = '2', s3 = '3'
      SET s4 = '4', s5 = '5', s6 = '6'
      SET s7 = '7', s8 = '8', s9 = '9'

      WHILE moves < 9 AND gameOver == FALSE DO
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// ================= PLAYER 1 TURN =================
SET p1Turn = TRUE
WHILE p1Turn == TRUE DO
   CLEAR Screen
   DISPLAY Header "ITEC 102: TIC-TAC-TOE"
   DISPLAY 3x3 Board (s1 to s9) and Match Metrics (n1, n2, scoreA, scoreB)

   PROMPT n1 + "'s turn (Press 1-9): "
   READ single key m

   IF m == '1' AND s1 != 'X' AND s1 != 'O' THEN
      SET s1 = 'X'
      SET p1Turn = FALSE
   ELSE IF m == '2' AND s2 != 'X' AND s2 != 'O' THEN
      SET s2 = 'X'
      SET p1Turn = FALSE
   ELSE IF m == '3' AND s3 != 'X' AND s3 != 'O' THEN
      SET s3 = 'X'
      SET p1Turn = FALSE
   ELSE IF m == '4' AND s4 != 'X' AND s4 != 'O' THEN
      SET s4 = 'X'
      SET p1Turn = FALSE
   ELSE IF m == '5' AND s5 != 'X' AND s5 != 'O' THEN
      SET s5 = 'X'
      SET p1Turn = FALSE
   ELSE IF m == '6' AND s6 != 'X' AND s6 != 'O' THEN
      SET s6 = 'X'
      SET p1Turn = FALSE
   ELSE IF m == '7' AND s7 != 'X' AND s7 != 'O' THEN
      SET s7 = 'X'
      SET p1Turn = FALSE
   ELSE IF m == '8' AND s8 != 'X' AND s8 != 'O' THEN
      SET s8 = 'X'
      SET p1Turn = FALSE
   ELSE IF m == '9' AND s9 != 'X' AND s9 != 'O' THEN
      SET s9 = 'X'
      SET p1Turn = FALSE
   ELSE
      DISPLAY "Invalid move or slot already taken. Press any key to try again."
      WAIT for key press
   END IF
END WHILE
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INCREMENT moves by 1

// Check Player 1 Win (Rows, Columns, Diagonals)
IF (s1 == 'X' AND s2 == 'X' AND s3 == 'X') OR
  (s4 == 'X' AND s5 == 'X' AND s6 == 'X') OR
  (s7 == 'X' AND s8 == 'X' AND s9 == 'X') OR
  (s1 == 'X' AND s4 == 'X' AND s7 == 'X') OR
  (s2 == 'X' AND s5 == 'X' AND s8 == 'X') OR
  (s3 == 'X' AND s6 == 'X' AND s9 == 'X') OR
  (s1 == 'X' AND s5 == 'X' AND s9 == 'X') OR
  (s3 == 'X' AND s5 == 'X' AND s7 == 'X') THEN
   INCREMENT scoreA by 1
   SET gameOver = TRUE
   CLEAR Screen
   DISPLAY Updated Board and Match Metrics
   DISPLAY "*** " + n1 + " WINS THIS ROUND! ***"
END IF

// Check Draw
IF moves == 9 AND gameOver == FALSE THEN
   SET gameOver = TRUE
   CLEAR Screen
   DISPLAY Updated Board and Match Metrics
   DISPLAY "=== IT'S A DRAW ROUND! ==="
END IF

IF gameOver == FALSE THEN
   // ================ PLAYER 2 TURN ================
   SET p2Turn = TRUE
   WHILE p2Turn == TRUE DO
      CLEAR Screen
      DISPLAY Header "ITEC 102: TIC-TAC-TOE"
      DISPLAY 3x3 Board (s1 to s9) and Match Metrics (n1, n2, scoreA, scoreB)

      PROMPT n2 + "'s turn (Press 1-9): "
      READ single key mv

      IF mv == '1' AND s1 != 'X' AND s1 != 'O' THEN
         SET s1 = 'O'
         SET p2Turn = FALSE
      ELSE IF mv == '2' AND s2 != 'X' AND s2 != 'O' THEN
         SET s2 = 'O'
         SET p2Turn = FALSE
      ELSE IF mv == '3' AND s3 != 'X' AND s3 != 'O' THEN
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            SET s3 = 'O'
            SET p2Turn = FALSE
        ELSE IF mv == '4' AND s4 != 'X' AND s4 != 'O' THEN
            SET s4 = 'O'
            SET p2Turn = FALSE
        ELSE IF mv == '5' AND s5 != 'X' AND s5 != 'O' THEN
            SET s5 = 'O'
            SET p2Turn = FALSE
        ELSE IF mv == '6' AND s6 != 'X' AND s6 != 'O' THEN
            SET s6 = 'O'
            SET p2Turn = FALSE
        ELSE IF mv == '7' AND s7 != 'X' AND s7 != 'O' THEN
            SET s7 = 'O'
            SET p2Turn = FALSE
        ELSE IF mv == '8' AND s8 != 'X' AND s8 != 'O' THEN
            SET s8 = 'O'
            SET p2Turn = FALSE
        ELSE IF mv == '9' AND s9 != 'X' AND s9 != 'O' THEN
            SET s9 = 'O'
            SET p2Turn = FALSE
        ELSE
            DISPLAY "Invalid move or slot already taken. Press any key to try again."
            WAIT for key press
        END IF
    END WHILE

    INCREMENT moves by 1

    // Check Player 2 Win (Rows, Columns, Diagonals)
    IF (s1 == 'O' AND s2 == 'O' AND s3 == 'O') OR
       (s4 == 'O' AND s5 == 'O' AND s6 == 'O') OR
       (s7 == 'O' AND s8 == 'O' AND s9 == 'O') OR
       (s1 == 'O' AND s4 == 'O' AND s7 == 'O') OR
       (s2 == 'O' AND s5 == 'O' AND s8 == 'O') OR
       (s3 == 'O' AND s6 == 'O' AND s9 == 'O') OR
       (s1 == 'O' AND s5 == 'O' AND s9 == 'O') OR
       (s3 == 'O' AND s5 == 'O' AND s7 == 'O') THEN
        INCREMENT scoreB by 1
        SET gameOver = TRUE
        CLEAR Screen
        DISPLAY Updated Board and Match Metrics
        DISPLAY "*** " + n2 + " WINS THIS ROUND! ***"
    END IF
END IF
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    END WHILE

    // ================= END OR RESTART MENU =================
    SET choiceMade = FALSE
    WHILE choiceMade == FALSE DO
      PROMPT "Press [1] End Game | [2] Restart Game: "
      READ single key choice

      IF choice == '1' THEN
        SET game = 0
        SET choiceMade = TRUE
        CLEAR Screen
        DISPLAY Header "FINAL SCORES"
        DISPLAY n1 + " (" + scoreA + ") | " + n2 + " (" + scoreB + ")"
        DISPLAY "Thank you for playing our application!"
      ELSE IF choice == '2' THEN
        SET choiceMade = TRUE
        DISPLAY "Game Restarted..."
      ELSE
        DISPLAY "Invalid choice. Please press either 1 or 2."
      END IF
    END WHILE

  END WHILE
END Main

END PROGRAM
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